K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(Autonomous)

SAMAYAPURAM, TRICHY-621 112



Incubating Minds,
Catalyzing Careers

DEPARTMENT OF INFORMATION TECHNOLOGY

ACTIVITY

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Register No : 2503811720521042

Course code : IT25042

Course name : Data Science and Statistical Analysis laboratory

Branch : B.Tech – Information Technology

Topic : Analyze global temperature change and climate patterns

Analyze global temperature change and climate patterns

Program :

```
# Global Temperature Change and Climate Patterns
# Data Science Project

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os

# 1. Load Dataset
df = pd.read_csv("annual-temperature-anomalies.csv")

print("Dataset Loaded Successfully!")
print("Dataset Shape:", df.shape)
print(df.head())

# 2. Display Dataset Columns
print("\nColumn Names:")
print(df.columns.tolist())

# 3. Select Global Temperature Data
global_df = df[
   df["Entity"].astype(str).str.contains(
      "World|Global", case=False, regex=True
```

```
    )
].copy()

print("\nGlobal Records:")
print(global_df.head())

# 4. Identify Columns
year_col = "Year"
temp_col = next(
    col for col in global_df.columns
    if "temp" in col.lower() or "anomal" in col.lower()
)

# 5. Clean Data
analysis = global_df[[year_col, temp_col]].copy()
analysis.columns = ["Year", "Temperature"]

analysis["Year"] = pd.to_numeric(
    analysis["Year"], errors="coerce"
)
analysis["Temperature"] = pd.to_numeric(
    analysis["Temperature"], errors="coerce"
)

analysis = analysis.dropna().sort_values("Year")

# 6. Statistical Analysis
print("\nGLOBAL TEMPERATURE ANALYSIS")
```

```
print("Total Records:", len(analysis))
print("First Year:", int(analysis["Year"].min()))
print("Last Year:", int(analysis["Year"].max()))
print("Mean:", round(analysis["Temperature"].mean(), 3))
print("Maximum:", round(analysis["Temperature"].max(), 3))
print("Minimum:", round(analysis["Temperature"].min(), 3))

# 7. Create Output Folder
os.makedirs("outputs", exist_ok=True)

# 8. Line Graph
plt.figure(figsize=(10, 5))
plt.plot(
  analysis["Year"],
  analysis["Temperature"],
  label="Temperature"
)
plt.title("Global Temperature Change Over Time")
plt.xlabel("Year")
plt.ylabel("Temperature Anomaly (°C)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("outputs/temperature_trend.png", dpi=300)
plt.show()

# 9. Decade-wise Analysis
analysis["Decade"] = (analysis["Year"] // 10) * 10
```

```
decade_avg = analysis.groupby("Decade")["Temperature"].mean()


plt.figure(figsize=(10, 5))
decade_avg.plot(kind="bar")
plt.title("Average Temperature by Decade")
plt.xlabel("Decade")
plt.ylabel("Temperature Anomaly (°C)")
plt.tight_layout()
plt.savefig("outputs/decade_comparison.png", dpi=300)
plt.show()

# 10. Save Results
analysis.to_csv("outputs/analysed_temperature.csv", index=False)

print("\nProject Completed Successfully!")
print("Graphs and analysed data saved in outputs folder.")
```

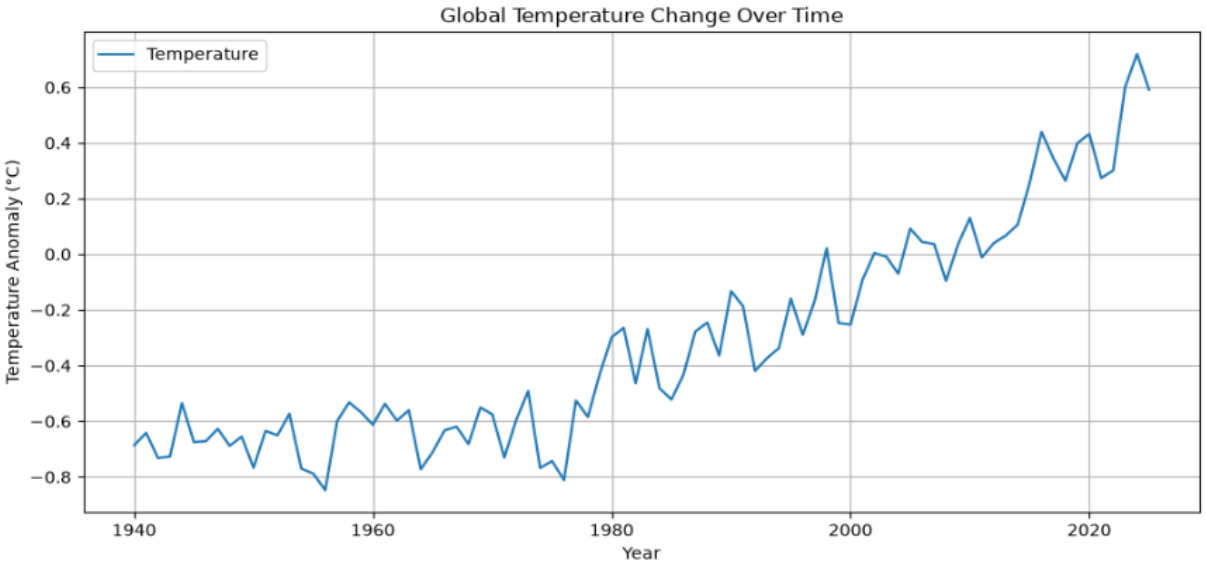
Output :

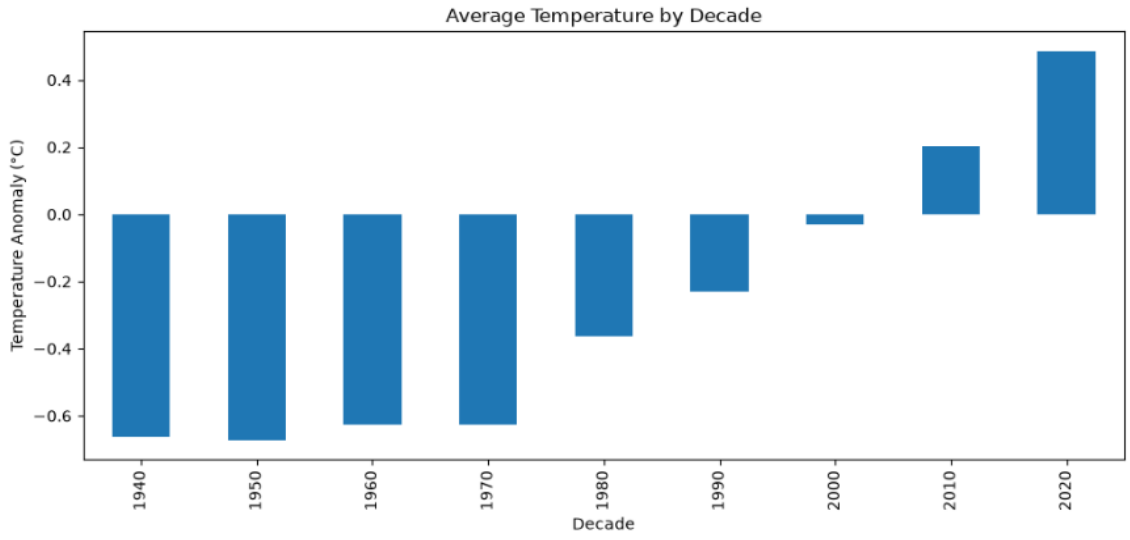
```
Dataset Loaded Successfully!
Dataset Shape: (18318, 4)
       Entity Code  Year  Temperature anomaly
0  Afghanistan  AFG  1940            -1.202316
1  Afghanistan  AFG  1941             0.794745
2  Afghanistan  AFG  1942             0.355437
3  Afghanistan  AFG  1943            -1.005242
4  Afghanistan  AFG  1944            -0.386346

Column Names:
['Entity', 'Code', 'Year', 'Temperature anomaly']

Global Records:
       Entity      Code  Year  Temperature anomaly
17974  World  OWID_WRL  1940            -0.686135
17975  World  OWID_WRL  1941            -0.641774
17976  World  OWID_WRL  1942            -0.732430
17977  World  OWID_WRL  1943            -0.726379
17978  World  OWID_WRL  1944            -0.534997

GLOBAL TEMPERATURE ANALYSIS
Total Records: 86
First Year: 1940
Last Year: 2025
Mean: -0.316
Maximum: 0.72
Minimum: -0.847
```






```
Project Completed Successfully!
Graphs and analysed data saved in outputs folder.
```